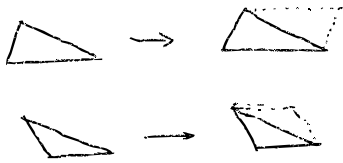
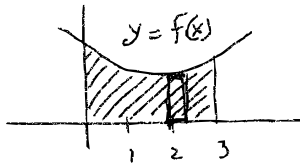


Exercise 4.1



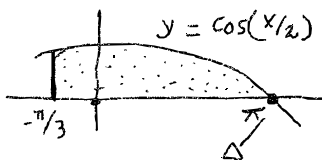
$$\text{so, } \Delta = \frac{1}{2} \square = \frac{1}{2} bh$$

Exercise 4.2.a

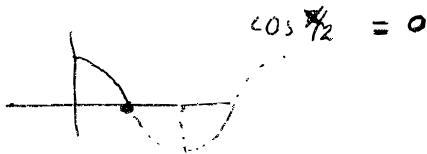


$$\int_0^3 f(x) dx$$

Exercise 4.2.b

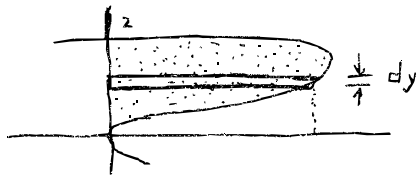


$$\int_{-\pi/3}^{\pi} \cos\left(\frac{x}{2}\right) dx$$



$$\cos \frac{\pi}{2} = 0$$

Exercise 4.2.c



$$\int_0^2 (-y^4 + y^3 + 2y^2) dy$$

Exercise 4.3.a

No, because $v(t)$ is always positive in the graph

Exercise 4.3.b

Yes, because during 'dt', $v(t)$ is constant. so $v(t) dt = \text{area of rectangle}$

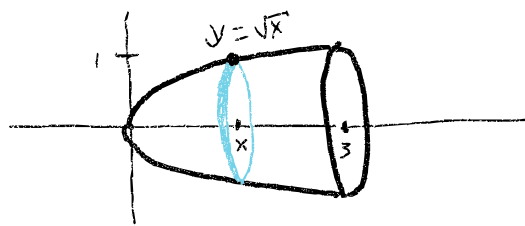


$$\int_4^8 v(t) dt$$

$$\int_{dt} \left\{ v(t) \right\} \boxed{v(t) dt}$$

↑
distance traveled
in dt

Exercise 4.4.a



Because $(x, \sqrt{x})$ will keep distance $\sqrt{x}$ from $(x, 0)$ during its 360° rotation around $(x, 0)$
That is a circle centered at $(x, 0)$ with radius $\sqrt{x}$

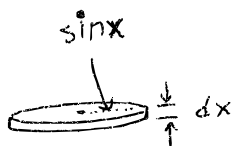
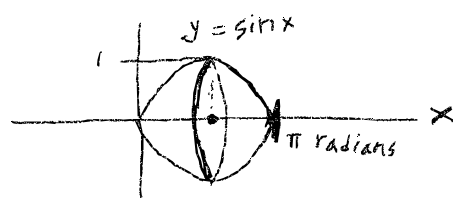
$$\int_0^3 \text{ (circle) } dx$$

$$\int_0^3 \pi (\sqrt{x})^2 dx$$

Exercise 4.4.b $\pi(\sqrt{x})^2 dx$: volume of a slice at x
 $= \pi x dx$

Exercise 4.4.c $\int_0^3 \pi (\sqrt{x})^2 dx$

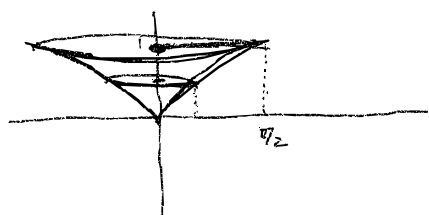
Exercise 4.4.d



volume of a slice = $\pi (\sin x)^2 dx$

$$\int_0^{\pi} \pi (\sin x)^2 dx$$

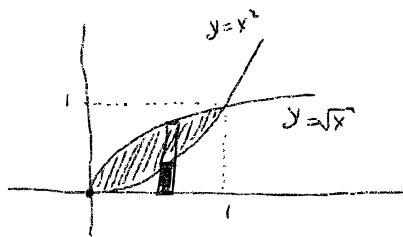
Exercise 4.4.e



$$\left. \begin{array}{l} y = \sin x \\ x = \sin^{-1} y \end{array} \right\}$$

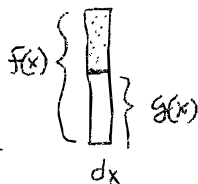
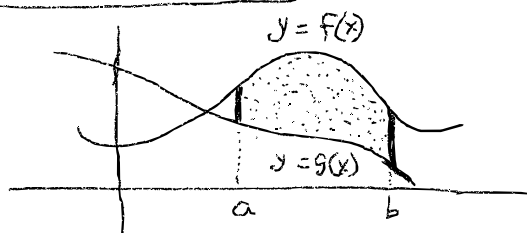
$$\int_0^1 \pi (\sin^{-1} y)^2 dy$$

Exercise 4.5.a



$$\int_0^1 (\sqrt{x} - x^2) dx$$

Exercise 4.5.b



$$\int_a^b (f(x) - g(x)) dx$$

Exercise 4.6

$\int_a^b f(x) dx = 6$ because in this interval $f(x)$ is never negative. So $f(x) dx$ is the area of ^{an} infinitely thin rectangular piece. Adding them all up gives the area under the curve between a and b .

Exercise 4.7.a

$$\int_0^a f(x) dx = -5$$

f

5

g

$$5 + 2 + 3 + 1 = 11$$

Exercise 4.7.b

$$\int_a^b f(x) dx = 2$$

Exercise 4.7.c

$$\int_0^b f(x) dx = -5 + 2 = -3$$

Exercise 4.7.d

$$\int_a^d f(x) dx = 2 - 3 + 1 = 0$$

Exercise 4.7.e

$$\int_0^d f(x) dx = -5 + 2 - 3 + 1 = -5$$

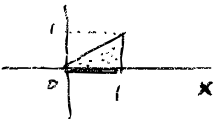
Exercise 4.8.a $\int_0^2 v(t) dt = \text{distance traveled in first two seconds}$

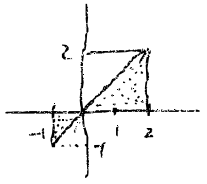
Ex 4.8.b $\int_0^5 v(t) dt = \text{displacement in first five seconds}$

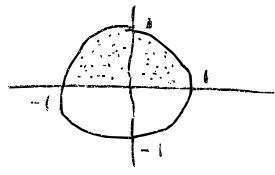
Ex 4.8.c $\int_0^5 |v(t)| dt = \text{distance traveled in first five sec.}$

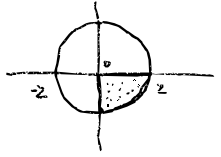
Ex 4.8.d $\int_0^2 v(t) dt = \int_0^2 |v(t)| dt \text{ (True)}$

Ex 4.8.e False because left distance > right distance

Ex 4.9.a $\int_0^1 x dx =$  $= \frac{1}{2}(1)(1) = \frac{1}{2} \text{ sq. unit}$

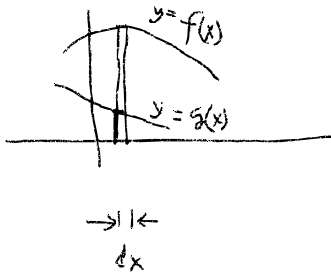
Ex 4.9.b $\int_{-1}^2 2x dx =$  $= 2 \left[\frac{1}{2}(2)(2) - \frac{1}{2}(1)(1) \right] = 2 \left[2 - \frac{1}{2} \right] = 3$

Ex 4.9.c $\int_{-1}^1 \sqrt{1-x^2} dx$ $y^2 = 1-x^2$ $y = \sqrt{1-x^2}$ 
 $= \frac{1}{2} \pi (1)^2 = \pi/2$

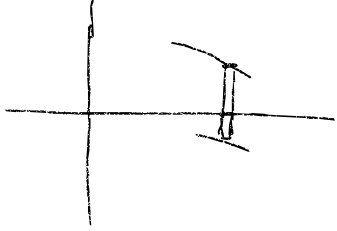
Ex 4.9.d $\int_0^2 -\sqrt{4-x^2} dx =$  $= -\frac{1}{4} \pi (2)^2 = -\pi$

Ex 4.9.e $\int_2^5 1 \cdot dx =$  $= 3$

Ex 4.10.a

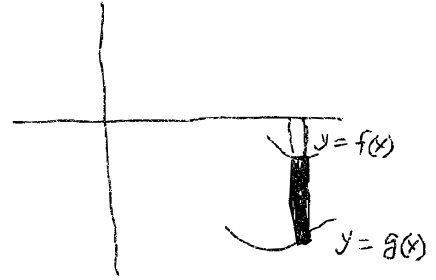


$$(f(x) - g(x)) dx$$



$$(f(x) + [-g(x)]) dx$$

to make it
a length



$$[-g(x) - (-f(x))] dx$$

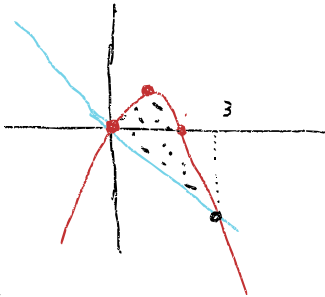
$$= [f(x) - g(x)] dx$$

Ex 4.10.b

Yes

$$\int_a^c [f(x) - g(x)] dx$$

Ex 4.10.c



$$-x = 1 - (x-1)^2$$

$$x^2 - 2x + 1 - x - 1 = 0$$

$$x^2 - 3x = 0$$

$$x(x-3) = 0$$

$$x = 0, 3$$

$$\int_0^3 [1 - (x-1)^2 + x] dx$$

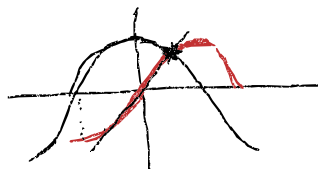
$$= \int_0^3 (1 - x^2 + 2x - 1 + x) dx$$

$$= \int_0^3 (3x - x^2) dx$$

Ex 4.11.a

even: $x^2, \cos x$

odd: $x, x^3, \sin x$

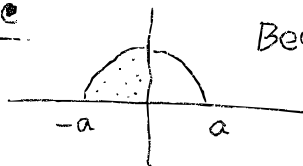


Ex 4.11.b

Even symmetric around $x=0$

Odd symmetric around $y=x$

Ex 4.11.c

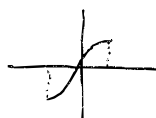


Because f is symmetric around $x=0$

So the same area on left and on right
of $(x=0)$.

Ex 4.11.d

Now same area on left and on right of $x=0$ but
sign changed.



Ex 4.11.e

$$f(-x) = -x^3 (\sin(x))^2 - (-x) \cos(-x)$$

(Odd)

$$= -x^3 \sin^2 x + x \cos x$$

$$= - (x^3 \sin^2 x - x \cos x)$$

$$= - f(x)$$

Ex 4.11.f

0

Ex 4.12.a

$$\frac{d}{dx} \left(c x^{p+1} /_{p+1} \right) = \frac{c}{p+1} (p+1) x^p = c x^p \quad \checkmark$$

Ex 4.12.b

$$x^3 \rightarrow \frac{1}{4} x^4$$

$$2x^9 \rightarrow 2 \cdot \frac{1}{10} x^{10}$$

$$x \rightarrow \frac{1}{2} x^2$$

$$-5\sqrt{x} \rightarrow -5 \frac{1}{3/2} x^{3/2} = \frac{10}{3} x^{3/2}$$

$$\sqrt[3]{x^2} = x^{2/3} \rightarrow \frac{1}{5/3} x^{5/3}$$

$$x^{-2} = \frac{1}{-1} x^{-1}$$

$$-8x^{-3/7} \rightarrow -8 \frac{1}{-3/7+1} x^{-3/7+1} = -\frac{56}{4} x^{4/7} = -14x^{4/7}$$

$$x^{-\pi} \rightarrow \frac{1}{1-\pi} x^{1-\pi}$$

Ex 4.12.c

$$c x^p \neq \frac{d}{dx} (c x^{p+1} /_{p+1}) \quad (p = -1) \quad \checkmark$$

$$c x^p \neq \frac{c}{p+1} \frac{d}{dx} (x^{p+1}) \quad c x^{-1} \rightarrow \frac{c}{0} x^0$$

Ex 4.13.a

$$\frac{d}{dx} F(x) = \cos x$$

↑
sin x

Ex 4.13.b

$$\frac{d}{dx} F(x) = -\sin x$$

↑
cos x

Ex 4.13.c

$$\frac{d}{dx} F(x) = \sin x$$

↑
-cos(x)

Ex 4.13.d

$$\frac{d}{dx} F(x) = \sec^2 x$$

↑
tan x

Ex 4.13.e

$$\frac{d}{dx} F(x) = e^x$$

↑
 e^x

Ex 4.13.f

$$\frac{d}{dx} F(x) = \sec x \tan x$$

↑
sec x

Ex 4.13.g

$$\frac{d}{dx} F(x) = \frac{1}{1+x^2}$$

↑
 $\tan^{-1}x$

Ex 4.13.h

$$\frac{d}{dx} F(x) = -\csc^2 x$$

↑
? $\cot x$

Ex 4.13.i

$$\frac{d}{dx} F(x) = \frac{1}{x}$$

↑
 $\ln x$

Ex 4.14.a

$$\frac{d}{dx} F(x) = \cos(5x)$$

↓
 $\frac{1}{5} \sin(5x)$

Ex 4.14.b

$$\frac{d}{dx} F(x) = \sin(\pi x)$$

↓
 $-\frac{1}{\pi} \cos(\pi x)$

Ex 4.14.c

$$\frac{d}{dx} F(x) = e^{-x}$$

↓
 $-e^{-x}$

Ex 4.14.d

$$\frac{d}{dx} F(x) = e^{3x}$$

↓
 $\frac{1}{3} e^{3x}$

Ex 4.14.e

$$\frac{d}{dx} F(x) = \sec^2\left(\frac{x}{2}\right)$$

↓
 $2 \tan\left(\frac{x}{2}\right)$

Ex 4.14.f

$$\frac{d}{dx} F(x) = 10^x$$

↓
 $\frac{1}{\ln(10)} 10^x$

Ex 4.14.g

$$\frac{d}{dx} F(x) = 5^{-x}$$

↓
 $-\frac{1}{\ln(5)} 5^{-x}$

Ex 4.14.h

$$\frac{d}{dx} F(x) = \frac{1}{1+4x^2} = \frac{1}{1+(2x)^2}$$

↓
 $\frac{1}{2} \tan^{-1}(2x)$

Ex 4.15.a

$$\frac{d}{dx} \ln x = \frac{1}{x}$$

defined for $x > 0$

$\ln(-x)$ is defined only for $x < 0$

$$\text{so } \frac{d}{dx} \ln(-x) = -\frac{1}{x} > 0$$

They have different domains.

Ex 4.15.b

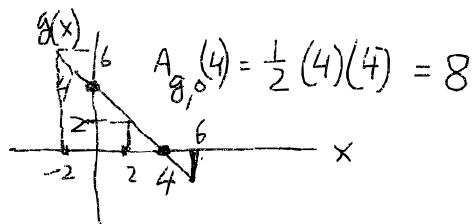
$g(x) = \frac{1}{x}$ has domain $x \in \mathbb{R} \setminus \{0\}$

$$G(x) = \begin{cases} \ln(x), & x > 0 \\ \ln(-x), & x < 0 \end{cases}$$

✓✓

$\ln(x)$

Ex 4.16.a



Ex 4.16.b

$$A_{g,2}(6) = 0$$

Ex 4.16.c

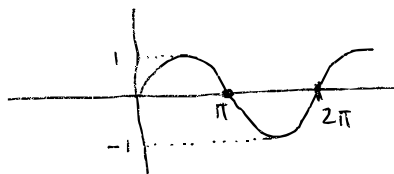
$$A_{g,-2}(4) = \frac{1}{2} \cdot 6 \cdot 6 = 18$$

Ex 4.16.d

$$A_{g,1}(1) = 0$$

Ex 4.17

$$s(x) = \sin x$$



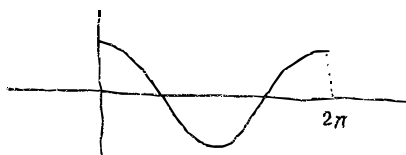
$$A_{s,0}(2\pi) = 0$$

$$A_{f,a}(x) = \int_a^x f(t) dt$$

Ex 4.18

$$g(x) = \cos(x)$$

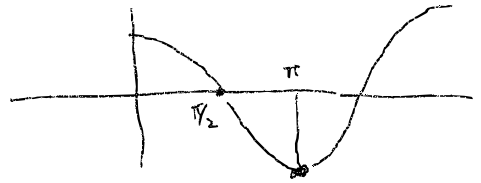
$$\begin{aligned} A_{g,0}(x) &= \int_0^x g(t) dt \\ &= \int_0^x \cos(t) dt \end{aligned}$$



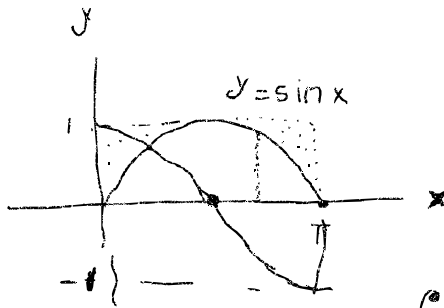
Fundamental theorem of calculus

$$f(x) \xrightarrow{\text{integrate}} A_{f,a}(x) (= \int_a^x f(t) dt)$$

differentiate



Example



$$A_{\sin,0}(x) = \int_0^x \sin(t) dt$$

$$A_{\sin,0}(\pi) = \int_0^{\pi} \sin(t) dt = ?$$

$$A_{\sin,0}(x) = -\cos(x) + C$$

$$A_{\sin,0}(0) = -\cos(0) + C = 0$$

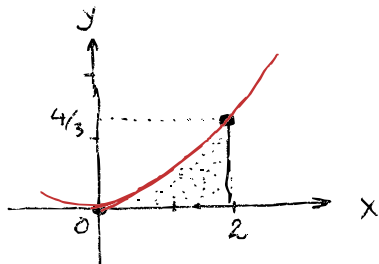
$$C = 1$$

$$A_{\sin,0}(x) = 1 - \cos(x)$$

$$A_{\sin,0}(\pi) = 1 - \cos(\pi) = 1 - (-1) = 2$$

Exercise 19

$$y = \frac{1}{3}x^2$$



$$A_{y,0}(x) = \int_0^x y(t) dt ; A_{y,0}(2) = \int_0^2 \frac{1}{3}x^2 dx$$

$$= \int_0^x \frac{1}{3}t^2 dt$$

$$A_{y,0}(x) = \frac{1}{9}x^3 + C$$

$$A_{y,0}(0) = C = 0$$

$$A_{y,0}(x) = \frac{1}{9}x^3$$

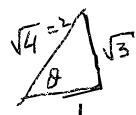
$$A_{y,0}(2) = \frac{1}{9} \cdot 2^3 = \frac{8}{9} \text{ unit}^2$$

$$A_{y,0}(0) = \tan^{-1}(0) + C = 1$$

$$\text{so } C = 0$$

$$A_{y,0}(x) = \tan^{-1} x$$

$$A_{y,0}(\sqrt{3}) = \tan^{-1}(\sqrt{3}) = \frac{\pi}{3} \text{ unit}^2$$



$$\tan^{-1}(\sqrt{3}) = \theta$$

$$\sqrt{3} = \tan \theta$$

Exercise 20

$$y = \frac{1}{1+x^2}$$

$$y = 1+x^2$$

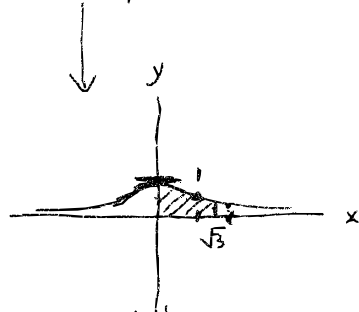
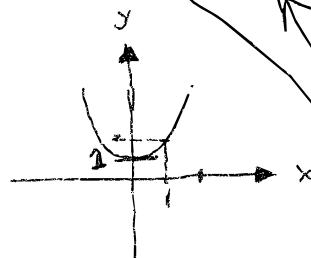
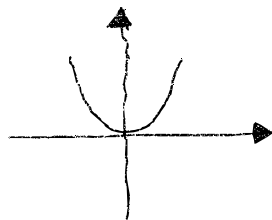
$$A_{y,0}(x) \equiv \text{antiderivative of } y = \frac{1}{1+x^2}$$

One antiderivative of y is $\tan^{-1}x$

$$\text{because } \frac{d}{dx}(\tan^{-1}x) = \frac{1}{1+x^2}$$

so all antiderivatives of y is like $\tan^{-1}x + C$

$$A_{y,0}(x) = \tan^{-1}x + C$$



Exercise 21

$$(a) \frac{d}{dx} \underbrace{\left(\int_a^x f(t) dt \right)}_{A_{f,0}(x)} = f(x)$$

$$(b) \frac{d}{dx} \left(\int_a^x \sin(t) dt \right) = \sin(x)$$

$$(c) \frac{d}{dx} \left(\int_a^x \sin(t^2) dt \right) = \sin(x^2)$$

$$(d) \frac{d}{dx} \left(\int_a^{x^2} \sin(t) dt \right) = \boxed{2x \sin(x^2)} \checkmark$$

$$y = A_{a,\sin}(x^2)$$

$$\underbrace{u = x^2} \quad \frac{dy}{dx} = \underbrace{\frac{dy}{du}}_{\sin(x^2)} \cdot \underbrace{\left(\frac{du}{dx} \right)}_{= 2x} = 2x$$

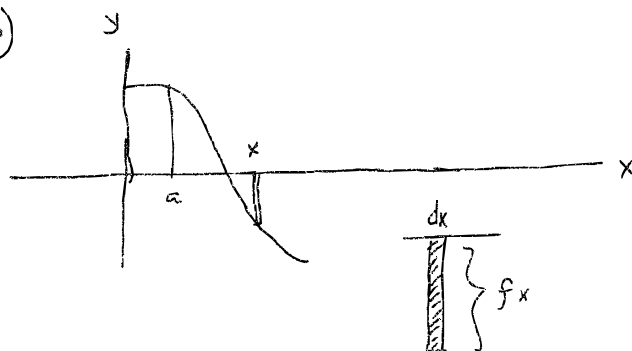
$$y = A_{a,\sin}(u)$$

$$\frac{d}{du} \left(\int_a^u \sin(t) dt \right) = \sin(u) = \sin x^2$$

Exercise 22

(a)

(b)

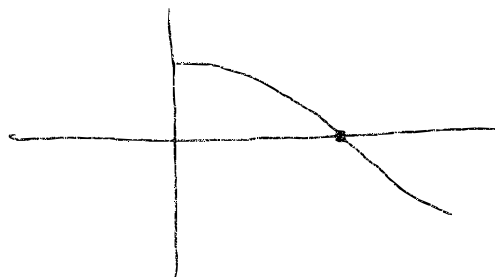


$$\Delta \text{ area} = - \underbrace{f(x) dx}$$

$$dA_{f,a}(x) = -(\Delta \text{ area}) = f(x) dx$$

$$\frac{dA_{f,a}(x)}{dx} = f(x)$$

(c)



even then it works because $dA_{f,a}(x) = 0$

which makes left side = 0 and right $f(x) = 0$ by defⁿ.

Chapter 4

(Example 3) $\int_0^2 (-2x^2 + 4x + 1) dx$

$$= \left[-2 \cdot \frac{1}{3} x^3 + 4 \cdot \frac{1}{2} x^2 + x \right]_0^2$$

$$= -\frac{2}{3} \cdot 2^3 + 2 \cdot 2^2 + 2$$

$$= -\frac{16}{3} + 8 + 2$$

$$= -\frac{16}{3} + 10$$

$$= \frac{30-16}{3}$$

$$= \frac{14}{3}$$

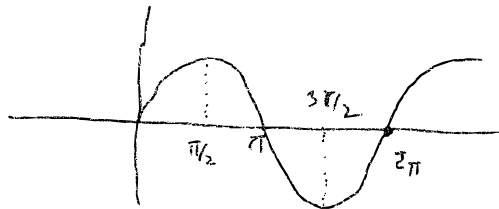
Exercise 4.23.a

$$\int_0^{3\pi/2} \cos x \, dx$$

$$= [\sin x]_0^{3\pi/2}$$

$$= \sin \frac{3\pi}{2} - 0$$

$$= -1$$



Exercise 4.23.b

$$\begin{aligned}\int_1^2 -2x^3 dx &= -2 \cdot \left[\frac{1}{4} x^4 \right]_1^2 \\ &= -\frac{1}{2} (2^4 - 1^4) \\ &= -\frac{1}{2} \cdot 15 \\ &= -15/2\end{aligned}$$

Exercise 4.23.c

$$\begin{aligned}\int_1^{e^2} \frac{1}{x} dx &= [\ln x]_1^{e^2} = \ln(e^2) - \ln(1) \\ &= 2 - 0 \\ &= 2\end{aligned}$$

Exercise 4.23.d

$$\begin{aligned}\int_1^{\sqrt{3}} \frac{1}{1+x^2} dx &= [\tan^{-1} x]_1^{\sqrt{3}} = \tan^{-1}(\sqrt{3}) - \tan^{-1}(1) \\ &= \frac{\pi}{3} - \frac{\pi}{4} \\ &= \frac{4\pi - 3\pi}{12} \\ &= \frac{\pi}{12}\end{aligned}$$

Exercise 4.23.e

$$\begin{aligned}\int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \sin x \, dx &= -[\cos x]_{\pi/6}^{\pi/4} \\&= -\left(\cos \frac{\pi}{4} - \cos \frac{\pi}{6}\right) \\&= \cos\left(\frac{\pi}{6}\right) - \cos\left(\frac{\pi}{4}\right) \\&= \frac{\sqrt{3}}{2} - \frac{\sqrt{2}}{2} \\&= \frac{\sqrt{3} - \sqrt{2}}{2}\end{aligned}$$

Exercise 4.23.f

$$\begin{aligned}\int_{-1}^8 \sqrt[3]{x} \, dx &= \frac{1}{1 + \frac{1}{3}} \left[x^{\frac{1}{3} + 1} \right]_{-1}^8 \\&= \frac{3}{4} \left(8^{\frac{4}{3}} - (-1)^{\frac{4}{3}} \right) \\&= \frac{3}{4} (16 - 1) \\&= \frac{45}{4}\end{aligned}$$

(Exercise 4.23.g)

$$\begin{aligned}\int_0^{\frac{\pi}{9}} \cos(3x) dx &= \left[\frac{1}{3} \sin(3x) \right]_0^{\frac{\pi}{9}} \\&= \frac{1}{3} \left(\sin \frac{\pi}{3} - \sin 0 \right) \\&= \frac{1}{3} \left(\frac{\sqrt{3}}{2} - 0 \right) \\&= \frac{\sqrt{3}}{6}\end{aligned}$$

(Exercise 4.23.h)

$$\begin{aligned}\int_0^{\frac{\sqrt{3}}{6}} \frac{1}{\sqrt{1-9x^2}} dx \\&= \frac{1}{3} \left[\sin^{-1}(3x) \right]_0^{\frac{\sqrt{3}}{6}} \\&= \frac{1}{3} \left[\sin^{-1}\left(\frac{\sqrt{3}}{2}\right) - 0 \right] \\&= \frac{1}{3} \cdot \frac{\pi}{3} \\&= \frac{\pi}{9}\end{aligned}$$

(Exercise 4.23.i)

$$\int_0^1 (x^9 + 5x^4 - 3x) dx$$

$$= \left[\frac{1}{10} x^{10} + 5 \cdot \frac{1}{5} x^5 - 3 \cdot \frac{1}{2} x^2 \right]_0^1$$

$$= \frac{1}{10} + 1 - \frac{3}{2}$$

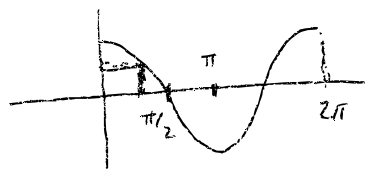
$$= \frac{1 + 10 - 15}{10}$$

$$= -\frac{4}{10}$$

$$= -\frac{2}{5}$$

(Exercise 4.23.j)

$$\int_0^{\pi/8} 2 \sin x \cos x \, dx = \int_0^{\pi/8} \sin 2x \, dx$$



$$= -\frac{1}{2} [\cos 2x]_0^{\pi/8}$$

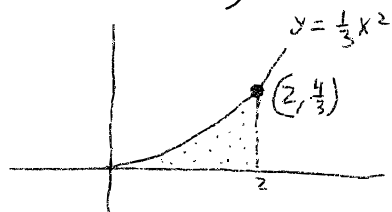
$$= -\frac{1}{2} \left(\cos \frac{\pi}{4} - 1 \right)$$

$$= -\frac{1}{2} \left(\frac{\sqrt{2}}{2} - 1 \right)$$

$$= -\frac{\sqrt{2}}{4} + \frac{1}{2}$$

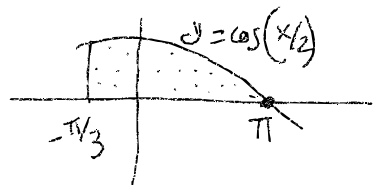
$$= \frac{2 - \sqrt{2}}{4}$$

(Exercise 4.24.a)



$$\text{area} = \int_0^2 \frac{1}{3} x^2 \, dx = \frac{1}{3} \cdot \frac{1}{3} [x^3]_0^2 = \frac{1}{9} (8 - 0) = \frac{8}{9}$$

(Exercise 4.24.b)

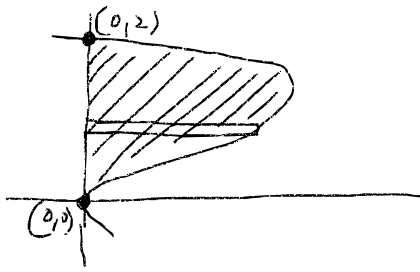


$$\text{area} = \int_{-\pi/3}^{\pi} \cos\left(\frac{x}{2}\right) \, dx = 2 \left[\sin\left(\frac{x}{2}\right) \right]_{-\pi/3}^{\pi}$$

$$= 2 \left[\sin\left(\frac{\pi}{2}\right) + \sin\left(\frac{\pi}{6}\right) \right] = 2 \left(1 + \frac{1}{2} \right)$$



(Exercise 4.24.c)



$$\text{area} = \int_0^2 (-y^4 + y^3 + 2y^2) dy$$

$$= \left[-\frac{1}{5}y^5 + \frac{1}{4}y^4 + \frac{2}{3}y^3 \right]_0^2$$

$$= -\frac{1}{5} \cdot 2^5 + \frac{1}{4} \cdot 2^4 + \frac{2}{3} \cdot 2^3$$

$$= -\frac{32}{5} + 4 + \frac{16}{3}$$

$$= \frac{-96 + 60 + 80}{15}$$

$$= \frac{44}{15}$$

$$-y^4 + y^3 + 2y^2 = 0$$

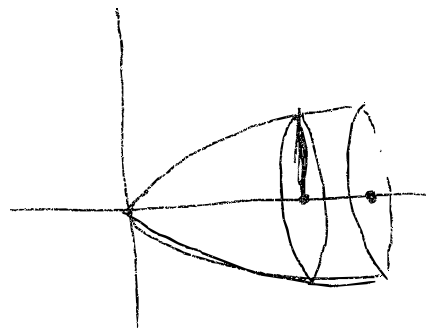
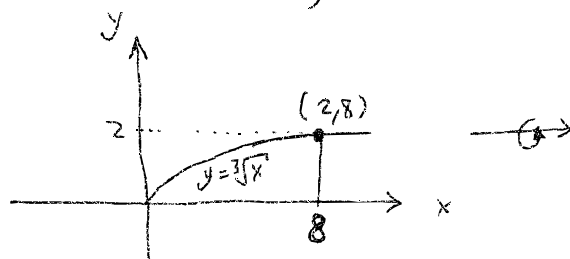
$$y^2 (-y^2 + y + 2) = 0$$



$$y^2 - y - 2 = 0$$

$$y = \frac{1 \pm \sqrt{1+8}}{2} = \frac{1 \pm 3}{2} = 2, -1$$

(Exercise 4.25)



$$\text{total volume} = \int_0^8 \pi x^{\frac{2}{3}} dx$$

$$= \frac{3\pi}{5} \left[x^{\frac{5}{3}} \right]_0^8$$

$$= \frac{3\pi}{5} \cdot 8^{\frac{5}{3}}$$

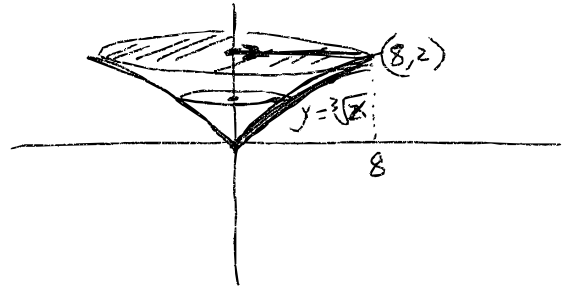
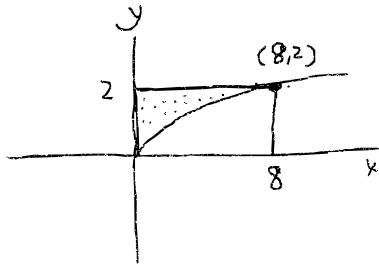
$$= \frac{3\pi}{5} \cdot 32$$

$$= \frac{96\pi}{5}$$

$$\text{area of slice} = \pi (\sqrt[3]{x})^2$$

$$\text{volume of slice} = \pi x^{\frac{2}{3}} dx$$

(Exercise 4.26)



$$y = \sqrt[3]{x}$$

$$y^3 = x$$

$$\begin{aligned} \text{slice area} &= \pi (y^3)^2 \\ &= \pi y^6 \end{aligned}$$

$$\text{volume of slice} = \pi y^6 dy$$

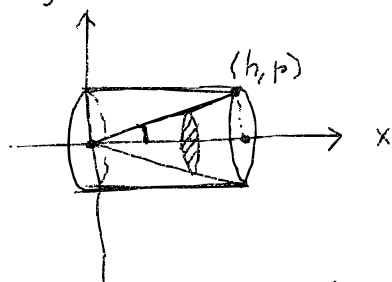
total volume

$$= \int_0^2 \pi y^6 dy$$

$$= \frac{\pi}{7} [y^7]_0^2$$

$$= \frac{\pi}{7} \cdot 128$$

(Exercise 4.27)



$$m = r/h$$

$$y = \frac{r}{h} x$$

$$\text{slice's area} = \pi \left(\frac{r}{h} x \right)^2$$

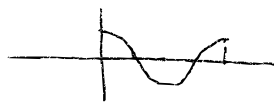
$$\text{slice's volume} = \frac{\pi r^2}{h^2} x dx$$

$$\text{cone's volume} = \int_0^h \frac{\pi r^2}{h^2} x^2 dx$$

$$= \frac{\pi r^2}{h^2} \cdot \frac{1}{3} [x^3]_0^h$$

$$= \frac{1}{3} \pi r^2 h$$

(Exercise 4.28)



$$\int_0^{\pi} \sin x dx = -[\cos x]_0^{\pi} = -\cos \pi + 1 = 2$$

$$\int_{\pi}^0 \sin x dx = -2$$

(Exercise 4.29)

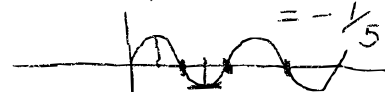
$$\begin{aligned} \int_a^b f(x) dx - \int_c^b f(x) dx + \int_c^d f(x) dx \\ = \int_a^b + \int_b^c + \int_c^d \\ = \int_a^d f(x) dx \end{aligned}$$

(Exercise 4.30)

(a) 2 (b) 3 (c) 6

(Exercise 4.31)

$$(a) \left[e F(x) \right]_a^b = e F(b) - e F(a) = e (F(b) - F(a)) = e [F(x)]_a^b$$

$$(b) \int_0^{3\pi/10} \cos(5x) dx = \frac{1}{5} [\sin(5x)]_0^{3\pi/10} = \frac{1}{5} \left(\sin \frac{3\pi}{2} - 0 \right) = \frac{1}{5} \sin\left(\frac{3}{2}\pi\right)$$


$= -\frac{1}{5}$

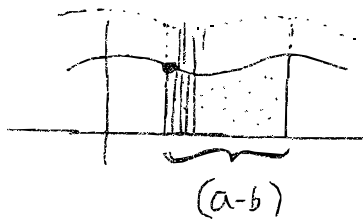
$$\int_0^1 e^{-5x} dx = -\frac{1}{5} [e^{-5x}]_0^1 = -\frac{1}{5} (e^{-5} - 1) = \frac{1}{5} (1 - e^{-5})$$

$$\int_4^{25} \sqrt{x} dx = \frac{2}{3} [x^{3/2}]_4^{25} = \frac{2}{3} (125 - 8) = \frac{234}{3} = 78$$

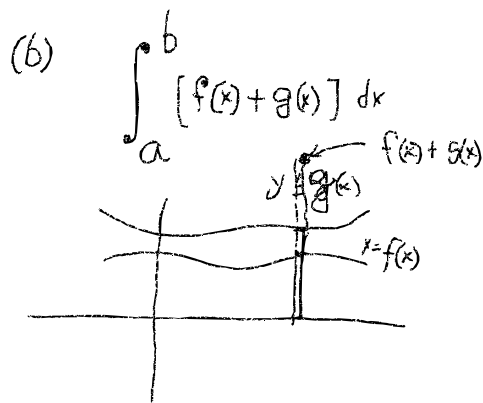
$$\begin{aligned} \int_0^{1/4} \frac{1}{1+16x^2} dx &= \frac{1}{4} [\tan^{-1}(4x)]_0^{1/4} = \frac{1}{4} (\tan^{-1}(1) - \tan^{-1}(0)) \\ &= \frac{1}{4} \left(\frac{\pi}{4} - 0 \right) = \frac{\pi}{16} \end{aligned}$$

(Exercise 4.32)

$$(a) \int_a^b c f(x) dx = c \int_a^b f(x) dx$$

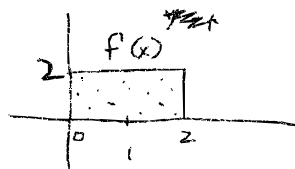


$$\int_a^b c f(x) dx = A_{c f(x), 0}(b) - A_{c f(x), 0}(a)$$

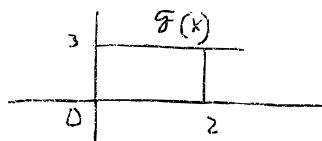


(Exercise 4.33)

$$\int_a^b f(x) g(x) dx \neq \left(\int_a^b f(x) dx \right) \left(\int_a^b g(x) dx \right)$$



$$\int_0^2 f(x) dx = 4$$



$$\int_0^2 g(x) dx = 6$$

$$\int_0^2 f(x) g(x) dx = 12 \neq \text{~~24~~ } (24)$$

(Exercise 4.34) If f is continuous over $[a, b]$...